

Unit - III

Activity Based Scheduling

* Introduction:

- Activity Based Scheduling is a project scheduling approach that focuses on identify, organizing and sequencing all individuals activities required to complete project.
 - Divides the project into a list of well-defined activities.
 - It provides a detailed view of "What needs to be done, when, and what order", ensuring efficient resource utilization and timely project completion.
 - A complete project plan, on the other hand must include a timetable detailing the start and finish timeframes for each task.
- We'll be able to do the following as a result of this:
- i) Ensure that the necessary resources are available at the precise moment needed;
 - ii) Prevent many activities for the same resource at same time;
 - iii) Create complete list of events who is performing each operation.
 - iv) Develop details plan to measure actual performance;
 - v) Make a cash flow prediction for a specific period of time;

* Objectives of Planning :-

- The objective of software project planning is to provide a framework that enables the manager to make reasonable estimates of resources, cost and schedule.

i) Feasibility Assessment :-

- A feasibility study is an assessment of feasibility of a proposed plan or project.
- It is not until we have constructed a detailed plan that we can forecast a completion date with any reasonable knowledge of its achievability.

ii) Resource Allocation :-

- Resource Allocation is the process of allocating and scheduling available resources as efficiently and economically as possible.
- Projects always need resources, but these resources are usually sacrificed.

iii) Detailed Costing :-

- After generating an activity plan and allocating particular resources, we will acquire extra complete estimates of prices and their timing.

iv) Motivation :-

- Providing targets and being seen to monitor achievements against target is an effective way of motivating staff, particularly where they have been involved in setting those targets in the first place.

v) Co-ordination :-

- The project plan offers a useful service means of communication and collaboration between team, especially for large projects with multiple project teams.
- Can be used as needed, will not be affected by forced shutdown.

vi) Risk Management :-

- By Breaking down the project into activities potential risks can be identified and managed at a more granular level.

vii) Time Management :-

- Activity planning helps in estimating the time required for each task and activity.

viii) Decision Making :-

- Activity planning provides a structured framework for making decisions throughout project lifecycle.

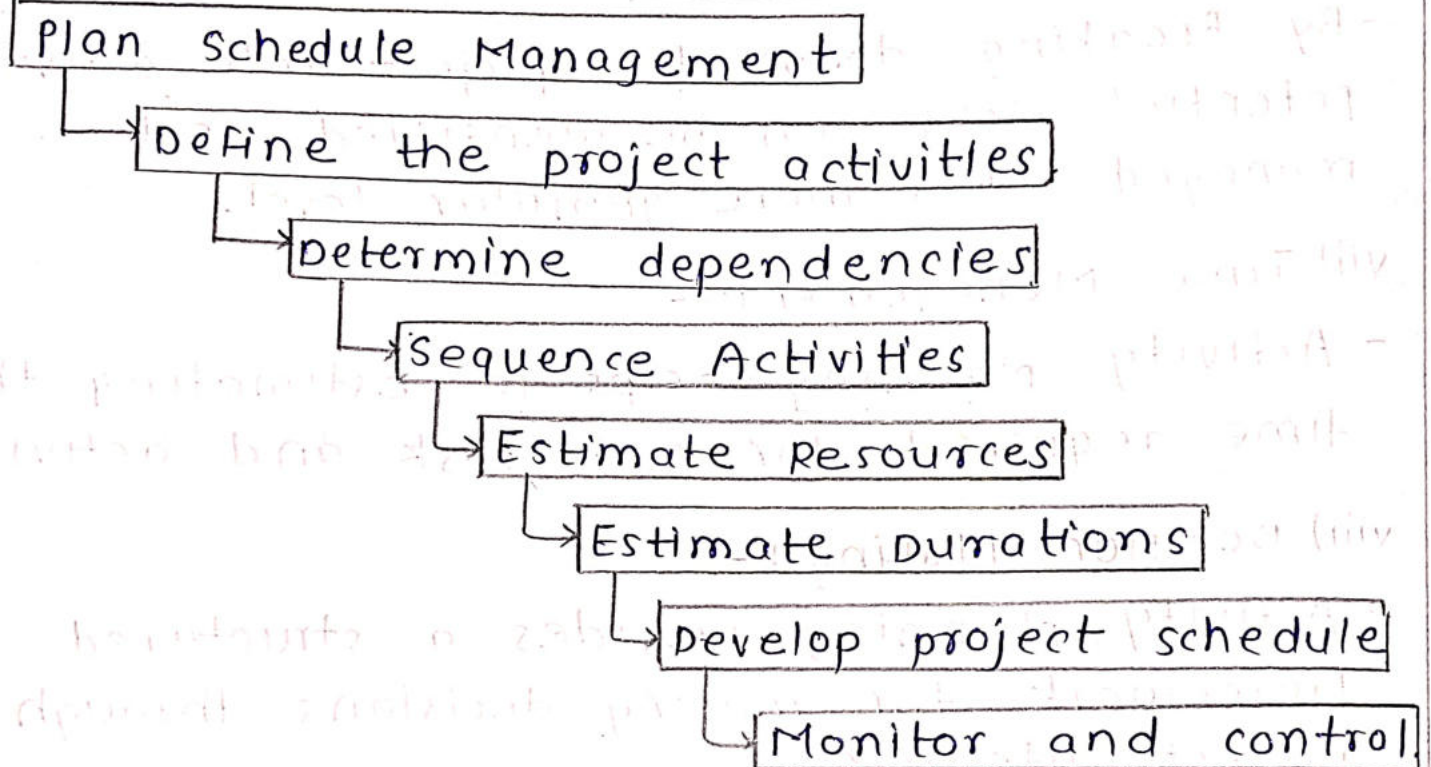
Que. Explain various steps involved and the objectives of activity planning in detail.

* Project Schedules :-

- A schedule in your project's time table actually consists of sequenced of activities and milestones that are needed to be delivered under a given period of time.

- Project Schedule simply means a mechanism that is used to communicate and know about that tasks are needed and has to be done or performed and which organizational resources will be given or allocated tasks and what time durations or time frame work is needed to be performed.

• The 8 steps of project scheduling process:-



• 4 phases to developing a project schedule:-

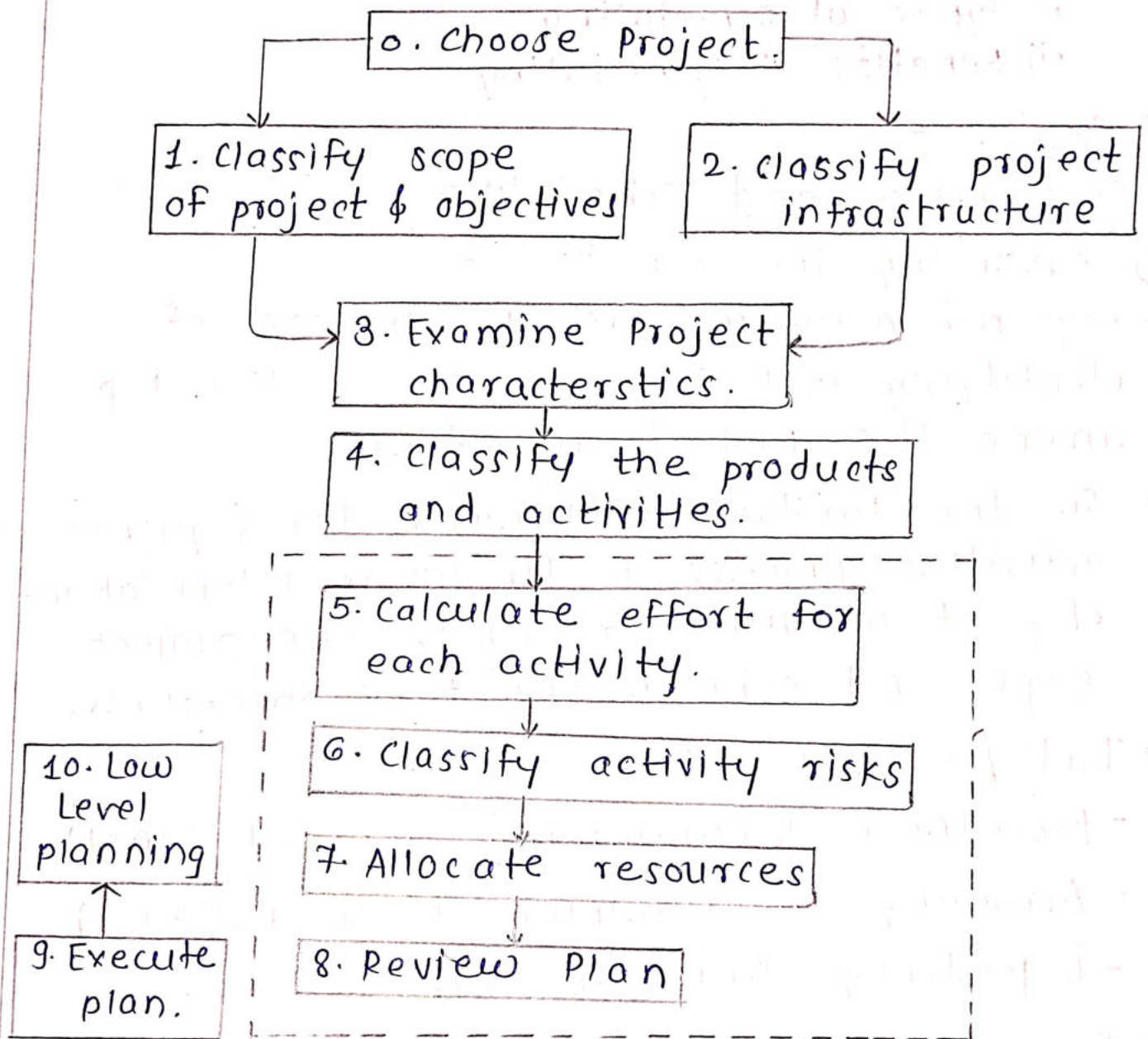
i) The initial stage in creating a plan is deciding which activities must be completed and in what sequence they must be completed. The activity plan is gradually created in steps 4 and 5 as shown figure.

ii) The second stage ideal activity plan then becomes subject of activity risk analysis,

the purpose of which is to identify potential problems that come up with an activity plan that will almost certainly affect allocation of resources.

iii) The third phase is to allocate resources.

iv) The last step is scheduling.



*Advantages:-

- i) Helps to identify relationship & monitor process.
- ii) Provides effective budget management.

* Types of scheduling:-

- 1) Project Management Scheduling
- 2) Operating System Scheduling
- 3) Production and Manufacturing Scheduling.

Que. Explain following in detail

- i) Scheduling
- ii) Types of scheduling
- iii) Benefits of scheduling.

Activities:-

* Sequencing and scheduling.

(I) Sequencing the activities:-

- Sequence activities are the process of identifying and documenting relationship among the project activities.
- So the fundamental reason for sequence activities process is finalizing interrelationship of activities to finish the project scope and achieve the task objectives.

• Tools / Techniques:

- Precedence Diagramming Method (PDM)
- Arrowing Diagramming Method (ADM)
- Dependency Determination

• Example: Activity A (collect data) → Activity B (analyze data) → Activity C (write report).

Here, sequencing ensures don't analyze before collecting.

II scheduling the activities:-

- The scheduling has had to take account of availability of staff and the way in which the activities have been allocated to them.
- Project Managers usually know the expected completion date of a project long before they plan and schedule the activities required to finish the project.
- To get your project finished on time, you need to schedule all the necessary project activities as quickly as possible.

• Tools / Techniques:-

- Gantt Charts
- Critical Path Method (CPM)
- Program Evaluation and Review Technique
- Resource Leveling & smoothing.
- Milestones charts.

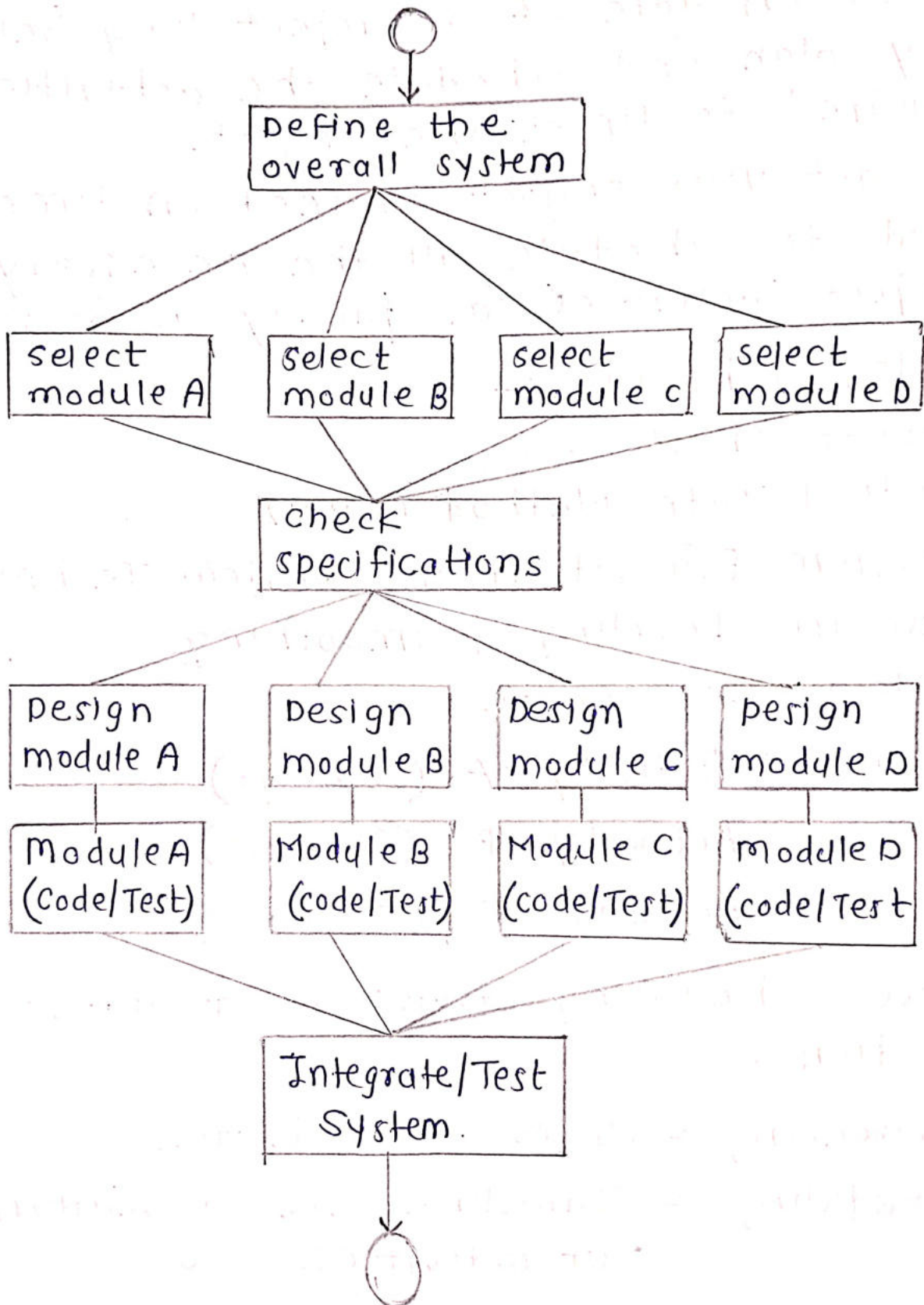
• Example: Activity A (2 days)
Activity B (3 days)
Activity C (2 days)

Hence, scheduling fixes exact dates and durations.

- Sequencing = Order of activities
- Scheduling = Timeline and allocation of activities.

* Network Planning Model.

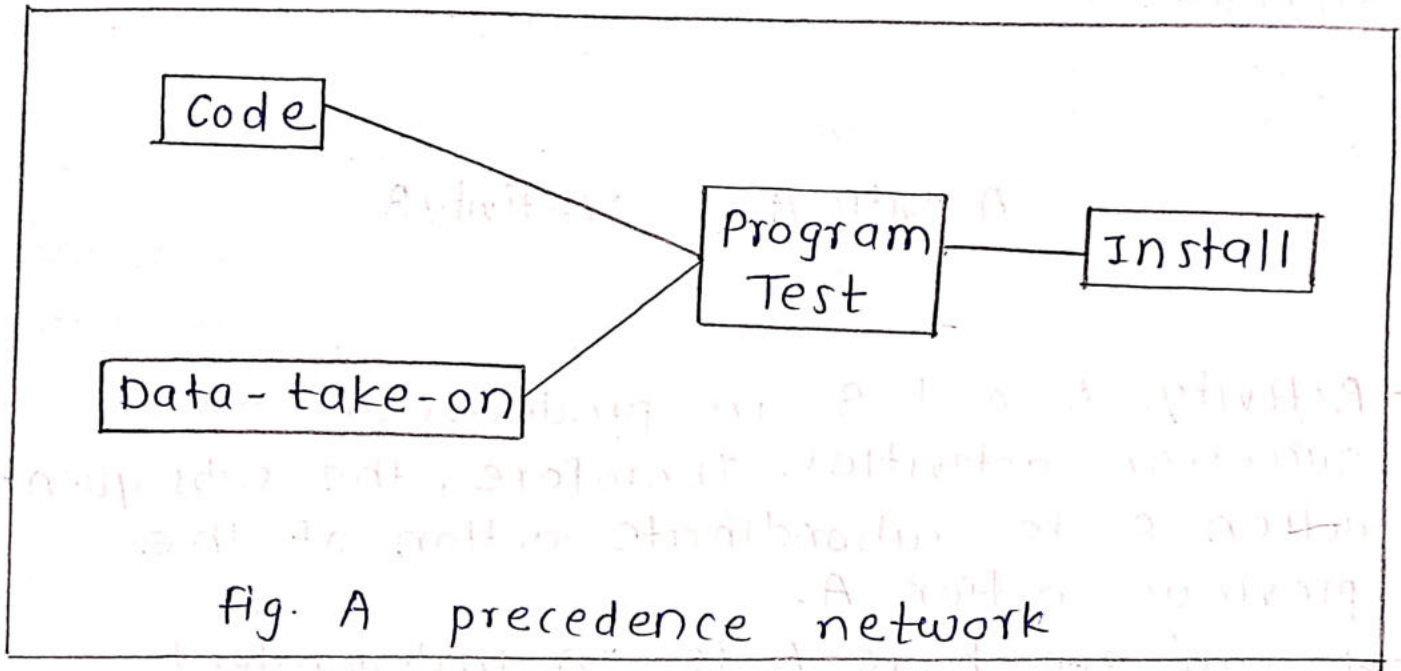
- Project activities and their interactions are modeled as networks in these project scheduling systems.
- Time moves from left to right in the network.



- Scheduling techniques model the projects activities and their relationships as a "Network".
- There are two best techniques:
 - 1) CPM (Critical Path Method)
 - 2) PERT (Program Evaluation & Review Technique)
- Both use the "Activity on Arrow" approach to visualize the project as a network where activities are drawn as arrows joining circles, or nodes which represent the possible start or completion on an activity or set of activities.
- Sequencing the tasks according to their logical relationship and then scheduling them taking into account resources and other factors.
- Most recently a variation on these techniques, called precedence networks, has become popular.
- This methods uses activity-on-node networks where activities are represented as nodes and the links between nodes represent precedence or sequencing requirements.

5) Links normally have no duration

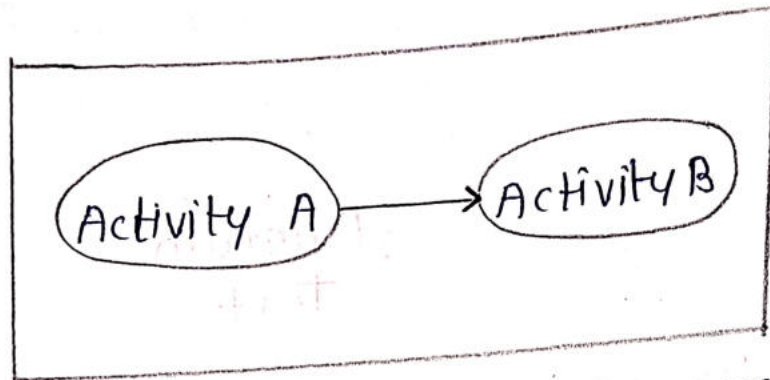
- Links represent the relationship between activities. Program testing cannot start until both coding and data take-on has been completed.



- In Fig. the "activity" "Program Test" cannot start until both "code" and "data take-on" have been completed and activity "install" cannot start until "Program Test" finished.
- 'Code' and 'data take-on' can therefore be said to be precedents of 'program test' and 'Program Test' precedent of 'install'.
- Note that we do not speak of 'code' and 'Data take-on' as precedents of 'install'.

* Activity Relationships (FS, SS, ST, FF)

- After determining the activities, we analyze them to see if there is any relationship between them.
- If there is a relationship, it is best to show it, so as a better with the project schedule.



- Activity A and B are predecessor and successor activities. Therefore, the subsequent action B is subordinate action of the previous action A.
- We can see here - A is an independent action.
- There are 4 types of relationships between Action A and B :-

1) Finish-to-start (FS)

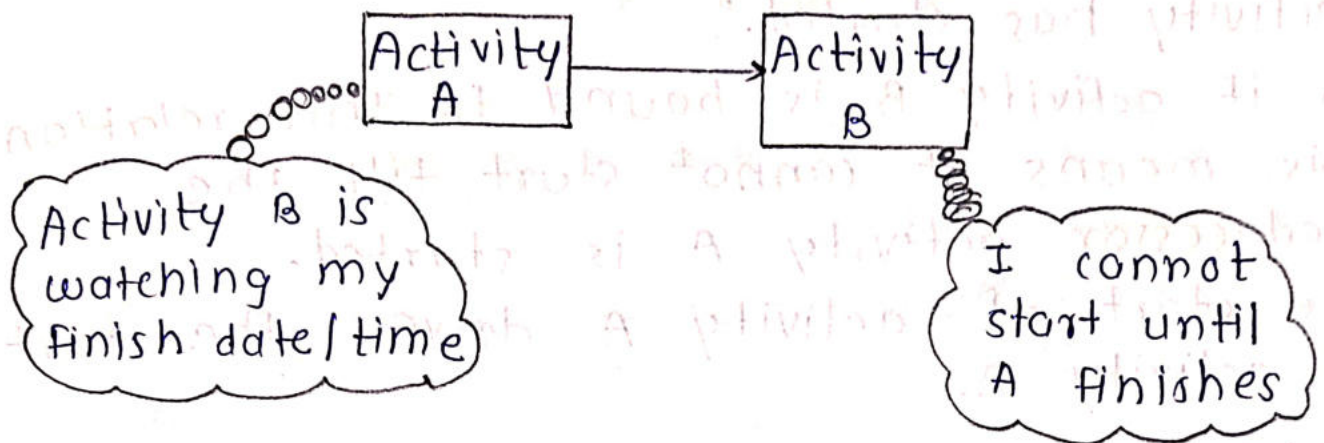
2) Finish-to-Finish (FF)

3) Start-to-start (SS)

4) Start-to-Finish (SF)

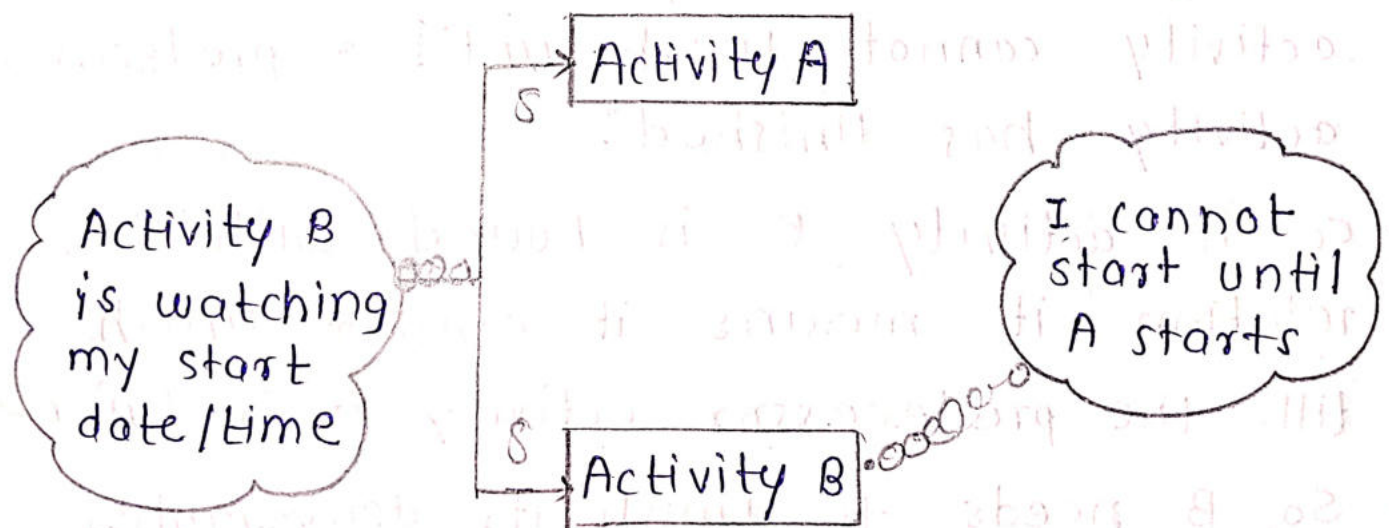
Explanation :-

1) Finish - to - start. (FS)



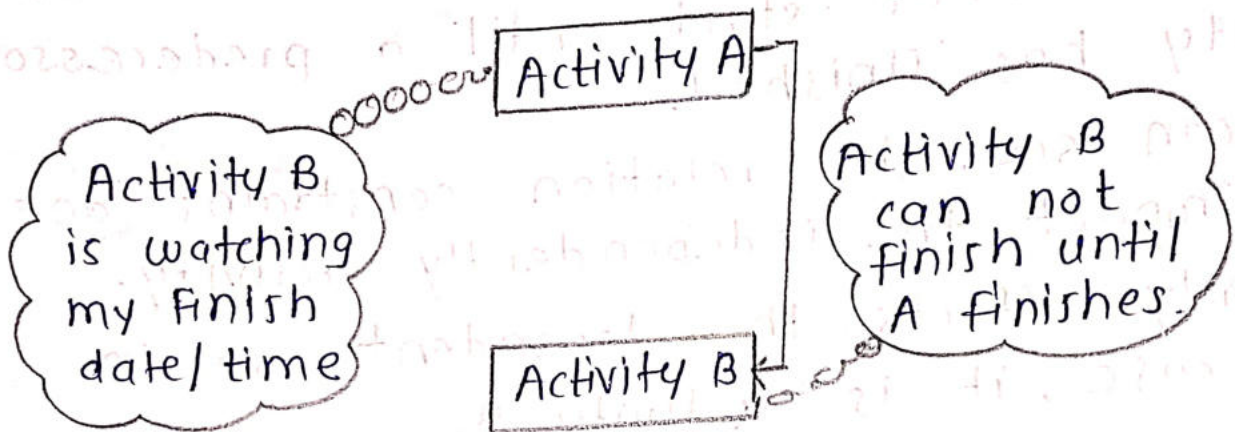
- "A logical connection in which a successor venture cannot start until a predecessor activity has finished."
- You can see, the relation constraint does not impact the independently activity.
- It only affects the dependent, so in our case, it is activity B.

2) Start - to - start. (SS)



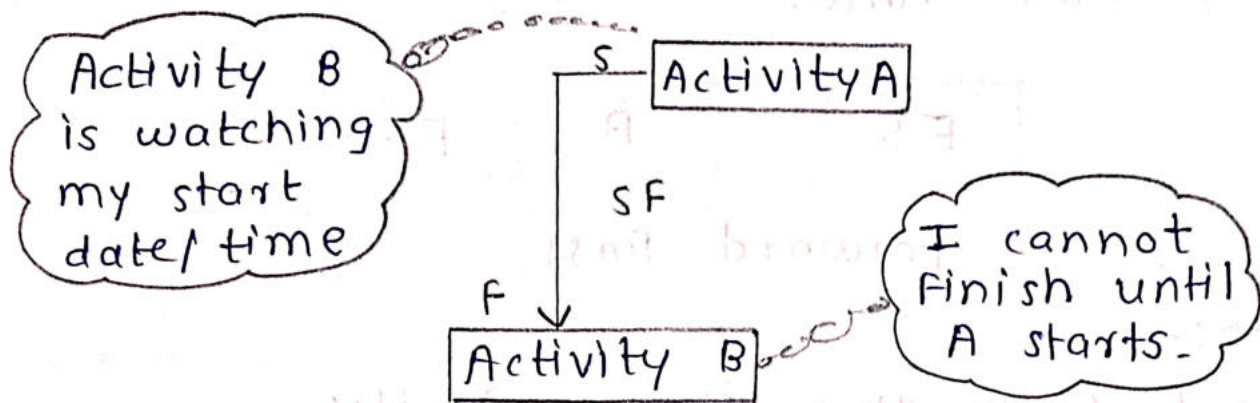
- "A logical relationship in which a successor activity cannot start until a predecessor activity has started."
- So if activity B is bound to this relation this means it cannot start till the predecessor activity A is started.
- The start of activity A drives the start of activity B.

3) Finish - to - Finish (FF)



- "A logical relationship in which a successor activity cannot finish until a predecessor activity has finished."
- So if activity B is bound with this relation it means it cannot finish till the predecessor activity A is finished.
- So, B needs to finish its deliverables and keep working with A till the time A is not done.

4) start-to-finish (SF)



- "A logical connection in which a successor activity cannot finish until a predecessor activity has started."
- So if activity B is bound to this relation in which it cannot finish till the independent activity A starts.

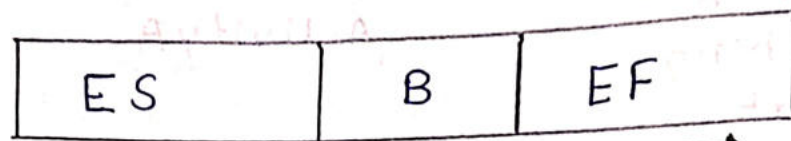
Que. Enlist and explain the 4 types of activity relationships?

* Forward Pass and Backward Pass technique:

i) Forward Pass Technique.

- Forward pass is a technique to move forward through network diagram to determine project duration and finding the critical path or free float of the project.
- Used to calculate Earliest start (ES) and Earliest finish (EF) times of each activity.

- Helps determine the minimum project completion time.



forward Pass.

• Steps:-

- i) start from the first activity.
- ii) Earliest start (ES):
 - for first activity, $ES = 0$.
 - for others, $ES = \text{Maximum of EF of all its predecessors}$.
- iii) Earliest Finish (EF)
 $EF = ES + \text{duration of activity}$.
- iv) Move forward node by node until the last node is reached.

• Formula:-

- $ES = \max(\text{EF all predecessors})$
- $EF = ES + \text{Duration}$.

ii) Backward Pass Technique:

- Backward pass represents moving backward to the end result to calculate late start or to find if there is any slack in the activity.
- Used to calculate Latest start (LS) and Latest Finish (LF) times of each activity.

- Helps determine the slack (float) and the critical Path.

LS	10	LF
----	----	----

← Backward Pass.

• Steps:-

- start from the last activity.
- Latest Finish (LF):
 - For the last activity, $LF = \text{Project completion time}$.
 - For others, $LF = \text{Minimum of LS of all its successors}$.
- Latest start (LS):
 - $LS = LF - \text{Duration of activity}$.
- Move backward node by node until the first node is reached.

• Formula:-

- $LF = \min (\text{LS of all successors})$
- $LS = LF - \text{Duration}$

Que. Explain following pass Technique:

- Forward Pass Technique
- Backward Pass Technique

* Critical Path and Remedies:

- The critical path in a project is the longest path of activities from start to finish, where:
 - Each activity has zero slack (Float).
 - Any delay in these activities will delay the whole project.
- It is found using Forward Pass and backward pass techniques.
- Float calculation:-
 - The tasks on the critical path have zero float, because they can't be delayed.

$$\text{Total Float} = LS - ES \text{ or } LF - EF$$

- Total Float shows the difference between the Earliest start (ES) and latest start (LS) of an activity before the completion date is delayed.

$$\text{Free Float} = \text{Lowest ES of successor} - EF$$

- Here Float represents the amount of time that an activity can be delayed before any successor activity will be delayed.

- Example :-

suppose a project has paths :-

• path 1 : $A \rightarrow B \rightarrow C \rightarrow D$ (18 days)

• path 2 : $E \rightarrow F \rightarrow G$ (14 days)

* Remedies for Critical Path Delays.

- When a project is at risk of delay due to issues in critical path activities, managers can apply the following remedies:

1) Crashing :- Add extra resources to shorten activity duration.

2) Fast Tracking :- Perform activities in parallel instead of sequential.

3) Reallocation of resources :- Shift resources from non-critical tasks to critical tasks.

4) Scope Reduction :- Reduce or simplify project requirements.

5) Improved Monitoring and control :-

Regularly track progress with tools like Gantt charts, CPM, PERT, or project management software.

Que. Discuss the network model represented by the CPM network.